

1.2.4 Reflection and Justification

Endangered Bird Species Analysis Visualisation

Reflection for VIS 1 (Seasonal Bar Chart)

Choice of Visualisation (Idiom)

A **100% Stacked Bar Chart** was implemented to analyze the relative distribution of endangered bird observations across the four Australian seasons. This idiom was selected because it facilitates part-to-whole comparisons between categorical groups (bird species) while normalizing for varying sample sizes. By scaling all bars to 100%, we eliminate the distraction of absolute counts and focus purely on seasonal proportions, which is essential for identifying migratory behavior.

Choice of Visual Variables and Layout

Position on a common scale (the y-axis) serves as the primary visual variable for magnitude, ensuring high precision in comparing percentages. **Color Hue** (using a qualitative ColorBrewer 'Set2' palette) is used to distinguish the four seasons. This provides clear categorical separation without implying an inherent ordering or importance between seasons.

Visual Hierarchy and Typography

The **Visual Hierarchy** is established by a clear title and legend. **Typography** uses a clean, sans-serif font for maximum legibility on screen. X-axis labels are rotated at a 45-degree angle to ensure that species names remain readable even as the number of categories increases, maintaining a tidy and professional layout.

Reflection for MAP (Interactive Leaflet Map)

Choice of Visualisation (Idiom)

An **Interactive Proportional Symbol Map** was selected for the spatial component. Since bird observations are inherently tied to geographic coordinates, mapping them is the only way to preserve spatial relationships. The interactive nature allows users to overcome the problem of overplotting through filtering, while the base map (OpenStreetMap) provides the necessary environmental context for the University staff audience.

Choice of Visual Variables and Layout

The primary variable is **Spatial Position**. **Size** (Area of the circle markers) is used to represent the density of sightings at specific coordinates. A square-root scaling was used for the radii to ensure that the area represents the data values linearly, avoiding perceptual distortion. **Color Hue** (Dark2 palette) is used as a categorical channel to differentiate species across the map.

Visual Hierarchy and Typography

The layout follows a functional hierarchy: controls (inputs) are grouped in a side panel, while the main visual (output) dominates the spatial layout. **Interactive Tooltips** provide a secondary level of information detail. The typography in the tooltips uses bold headers for 'Common Name' and 'Count', allowing users to extract key insights quickly during the exploration process.